

2.1 Data visualization (using seaborn, matplotlib and bokeh)**2.1.1 Working with Scatter plot, line char, bar chart and histogram using matplotlib, seaborn and bokeh****Data Visualization using Matplotlib in Python**

Matplotlib is a widely-used Python library used for creating static, animated and interactive data visualizations. It is built on the top of **NumPy** and it can easily handles large datasets for creating various types of plots such as line charts, bar charts, scatter plots, etc. These visualizations help us to understand data better by presenting it clearly through graphs and charts. In this article, we will see how to create different types of plots and customize them in matplotlib.

pip install matplotlib

1. Line Chart

Line chart is one of the basic plots and can be created using the **plot()** function. It is used to represent a relationship between two data X and Y on a different axis.

Syntax:

matplotlib.pyplot.plot(x, y, color=None, linestyle='-', marker=None, linewidth=None, markersize=None)

2. Bar Chart

A bar chart is a graph that represents the category of data with rectangular bars with lengths and heights which is proportional to the values which they represent. The bar plot can be plotted horizontally or vertically. It describes the comparisons between different categories and can be created using the **bar()** method.

Syntax:

matplotlib.pyplot.bar(x, height, width=0.8, bottom=None, color=None, edgecolor=None, linewidth=None)

3. Histogram

A histogram is used to represent data provided in a form of some groups. It is a type of bar plot where the X-axis represents the bin ranges while the Y-axis gives information about frequency. The **hist()** function is used to find and create histogram of x.

Syntax:

matplotlib.pyplot.hist(x, bins=None, range=None, density=False, color=None, edgecolor=None, alpha=None)

4. Scatter Plot

Scatter plots are used to observe relationships between variables. The **scatter()** method in the matplotlib library is used to draw a scatter plot.

Syntax:

matplotlib.pyplot.scatter(x, y, s=None, c=None, marker=None, linewidths=None, edgecolors=None, alpha=None)

Chart Type	Best For	Function Used
Scatter Plot	Correlation between two variables	plt.scatter()
Line Chart	Trends or progress over time	plt.plot()
Bar Chart	Comparing values across categories	plt.bar()
Histogram	Distribution/Frequency of a single value	plt.hist()

```
import matplotlib.pyplot as plt

# Sample Data
# Scatter and Line Chart Data
hours = [1, 2, 3, 4, 5]
marks = [40, 45, 50, 60, 70]

# Bar Chart Data
products = ['A', 'B', 'C', 'D']
sales = [250, 400, 150, 300]

# Histogram Data
student_marks = [45, 55, 67, 70, 85, 89, 70, 55, 60, 75, 90, 95, 62, 67, 73]

# Create a 2x2 subplot layout
fig, axs = plt.subplots(2, 2, figsize=(12, 10))
fig.suptitle('Matplotlib Visualization Examples', fontsize=16)

# 1. Scatter Plot
axs[0, 0].scatter(hours, marks, color='blue', marker='o')
axs[0, 0].set_title('Scatter Plot: Hours vs Marks')
axs[0, 0].set_xlabel('Hours Studied')
axs[0, 0].set_ylabel('Marks Scored')
axs[0, 0].grid(True)

# 2. Line Chart
days = ['Mon', 'Tue', 'Wed', 'Thu', 'Fri']
temperature = [30, 32, 33, 31, 29]
axs[0, 1].plot(days, temperature, color='green', marker='s', linestyle='--')
axs[0, 1].set_title('Line Chart: Temperature over Days')
axs[0, 1].set_xlabel('Day')
axs[0, 1].set_ylabel('Temperature (°C)')
axs[0, 1].grid(True)
```

3. Bar Chart

```

axs[1, 0].bar(products, sales, color='orange')
axs[1, 0].set_title('Bar Chart: Product Sales')
axs[1, 0].set_xlabel('Product')
axs[1, 0].set_ylabel('Sales')

```

4. Histogram

```

axs[1, 1].hist(student_marks, bins=5, color='purple', edgecolor='black')
axs[1, 1].set_title('Histogram: Marks Distribution')
axs[1, 1].set_xlabel('Marks Range')
axs[1, 1].set_ylabel('Number of Students')
axs[1, 1].grid(True)

```

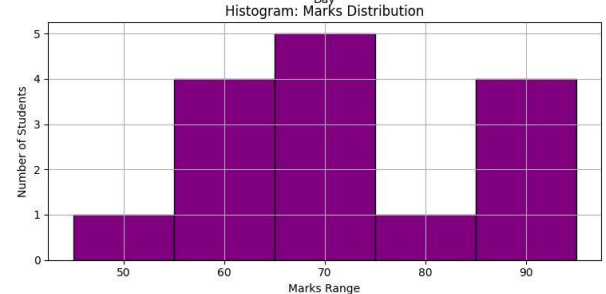
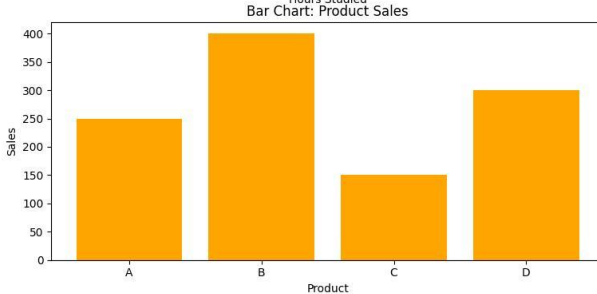
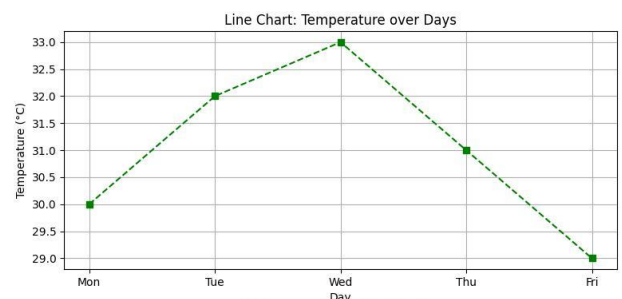
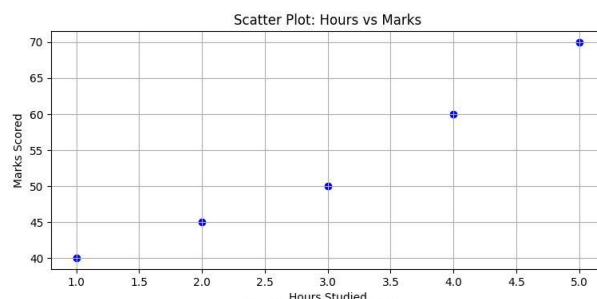
Adjust layout

```

plt.tight_layout(rect=[0, 0, 1, 0.96])
plt.show()

```

Matplotlib Visualization Examples



Available Seaborn Styles

Style Name	Description
"white"	Plain white background, no gridlines — useful for clean plots
"dark"	Dark background with gridlines — good for presentations
"whitegrid"	White background with light gridlines — great for statistical plots
"darkgrid"	Dark background with gridlines — best for contrasting visuals
"ticks"	White background with ticks on axes — elegant and professional look

```

import seaborn as sns
import matplotlib.pyplot as plt

```

```
# Use Seaborn style
sns.set(style="darkgrid")
#sns.set(style="ticks")

# Data
# Scatter & Line Chart
hours = [1, 2, 3, 4, 5]
marks = [40, 45, 50, 60, 70]

# Line Chart
days = ['Mon', 'Tue', 'Wed', 'Thu', 'Fri']
temperature = [30, 32, 33, 31, 29]

# Bar Chart
products = ['A', 'B', 'C', 'D']
sales = [250, 400, 150, 300]

# Histogram
student_marks = [45, 55, 67, 70, 85, 89, 70, 55, 60, 75, 90, 95, 62, 67, 73]

# Create 2x2 subplot layout
fig, axs = plt.subplots(2, 2, figsize=(14, 10))
fig.suptitle('Seaborn & Matplotlib Combined Charts', fontsize=18)

# 1. Scatter Plot
sns.scatterplot(x=hours, y=marks, ax=axs[0, 0], color='blue', s=100)
axs[0, 0].set_title('Scatter Plot: Hours vs Marks')
axs[0, 0].set_xlabel('Hours Studied')
axs[0, 0].set_ylabel('Marks Scored')

# 2. Line Chart
sns.lineplot(x=days, y=temperature, ax=axs[0, 1], color='green', marker='o')
axs[0, 1].set_title('Line Chart: Temperature over Days')
axs[0, 1].set_xlabel('Day')
axs[0, 1].set_ylabel('Temperature (°C)')

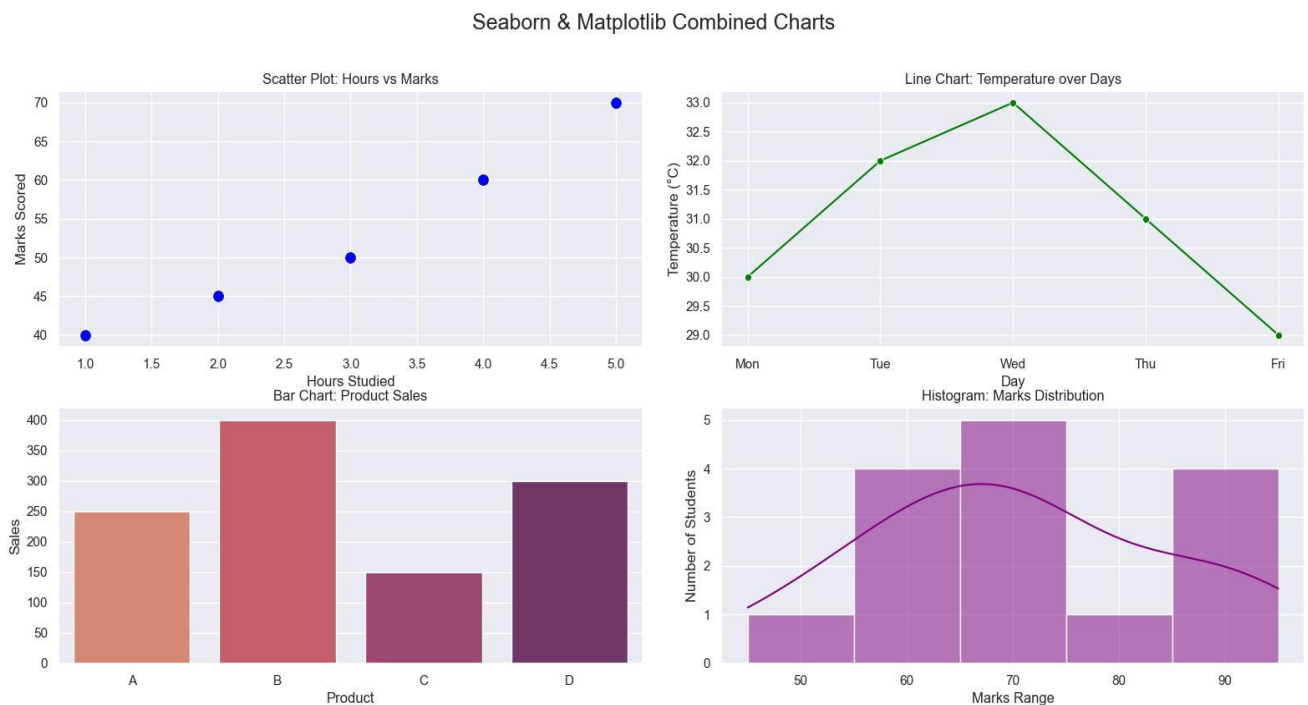
# 3. Bar Chart
sns.barplot(x=products, y=sales, ax=axs[1, 0], palette='flare')
axs[1, 0].set_title('Bar Chart: Product Sales')
axs[1, 0].set_xlabel('Product')
axs[1, 0].set_ylabel('Sales')

# 4. Histogram
sns.histplot(student_marks, bins=5, kde=True, ax=axs[1, 1], color='purple')
axs[1, 1].set_title('Histogram: Marks Distribution')
axs[1, 1].set_xlabel('Marks Range')
axs[1, 1].set_ylabel('Number of Students')
```

```
# Improve layout
plt.tight_layout(rect=[0, 0, 1, 0.95])

# Save figure
plt.savefig('combined_charts.png', dpi=300)

# Show plot
plt.show()
```



Bokeh is a data visualization library in Python that provides high-performance interactive charts and plots. Bokeh output can be obtained in various mediums like notebook, html and server. It is possible to embed bokeh plots in Django and flask apps.

```
pip install bokeh
```

Bokeh provides two visualization interfaces to users:

bokeh.models : A low level interface that provides high flexibility to application developers. **bokeh.plotting** : A high level interface for creating visual glyphs.

```
from bokeh.plotting import figure, show, output_file
from bokeh.layouts import gridplot
from bokeh.models import ColumnDataSource
import numpy as np
```

```

# Output file
output_file("bokeh_all_charts.html")

# -----
# 1. Scatter Plot: Hours vs Marks
# -----
hours = [1, 2, 3, 4, 5]
marks = [40, 45, 50, 60, 70]

p1 = figure(title="Scatter Plot: Hours vs Marks",
            x_axis_label='Hours Studied', y_axis_label='Marks Scored',
            width=400, height=300)
p1.circle(hours, marks, size=10, color="blue", alpha=0.7)

# -----
# 2. Line Chart: Temperature over Days
# -----
days = ['Mon', 'Tue', 'Wed', 'Thu', 'Fri']
temperature = [30, 32, 33, 31, 29]

p2 = figure(title="Line Chart: Temperature over Days",
            x_range=days,
            width=400, height=300)
p2.line(days, temperature, line_width=2, color="green")
p2.circle(days, temperature, size=8, color="green")

# -----
# 3. Bar Chart: Product Sales
# -----
products = ['A', 'B', 'C', 'D']
sales = [250, 400, 150, 300]
source = ColumnDataSource(data=dict(products=products, sales=sales))

p3 = figure(title="Bar Chart: Product Sales",
            x_range=products, y_axis_label='Sales',
            width=400, height=300)
p3.vbar(x='products', top='sales', width=0.5, source=source, color="orange")

# -----
# 4. Histogram: Marks Distribution
# -----
student_marks = [45, 55, 67, 70, 85, 89, 70, 55, 60, 75, 90, 95, 62, 67, 73]
hist, edges = np.histogram(student_marks, bins=5)

p4 = figure(title='Histogram: Marks Distribution',
            x_axis_label='Marks Range', y_axis_label='No. of Students',
            width=400, height=300)
p4.quad(top=hist, bottom=0, left=edges[:-1], right=edges[1:],

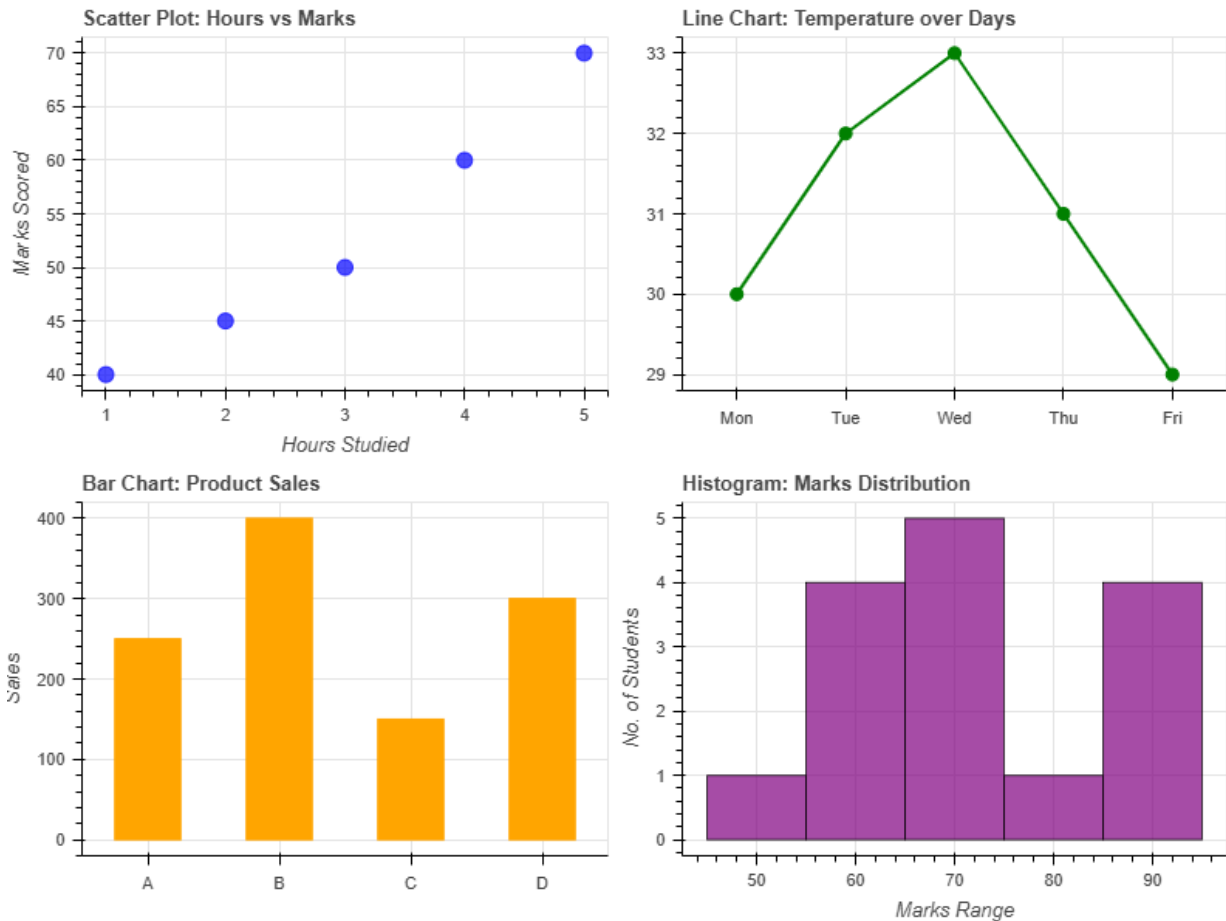
```

```

fill_color="purple", line_color="black", alpha=0.7)

# -----
# Layout and Display
# -----
layout = gridplot([p1, p2], [p3, p4])
show(layout)

```

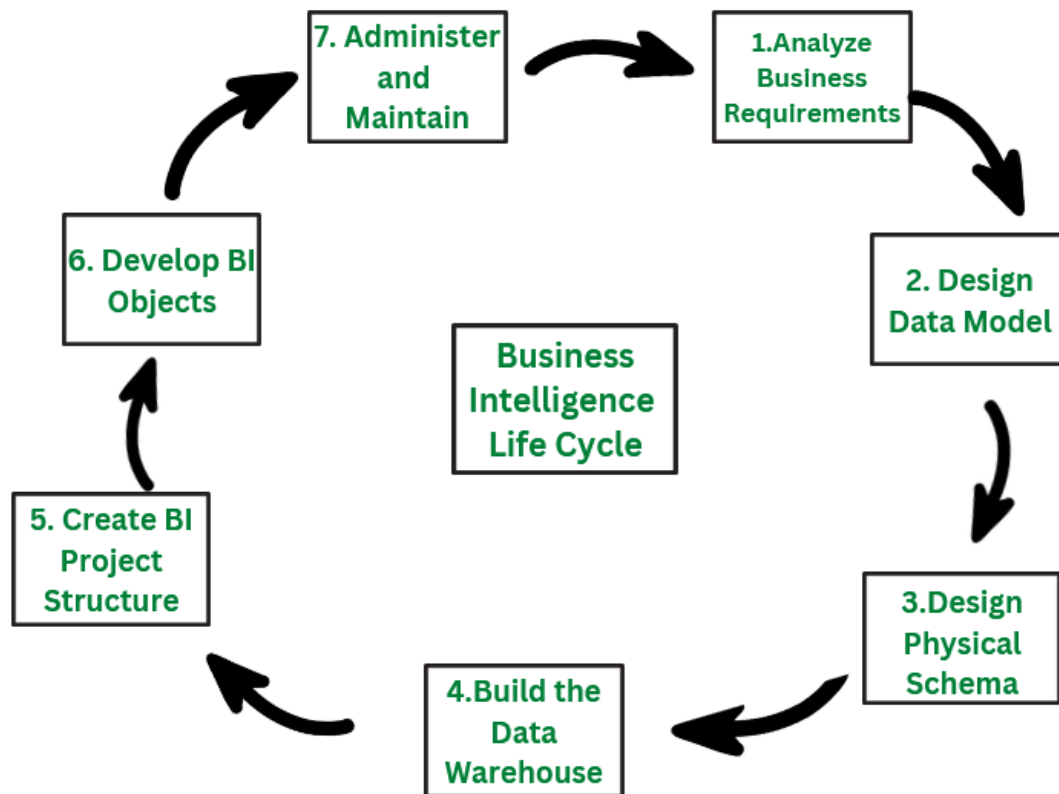


What is Business Intelligence?

Business Intelligence is the talk of a new changing and growing world that can be defined as a set of concepts and methodologies to improve decision-making in business through the use of facts and fact-based systems. The **Goal of Business Intelligence** is to improve decision-making in business ideas and analysis. Business Intelligence is not just a concept; it's a group of concepts and methodologies. Business Intelligence uses analytics and gut feelings for making decisions.

Business Intelligence is the process of analyzing unprocessed data and turning it into knowledge that the company can use to make decisions. Organizations can better understand their marketing strategies, earnings and losses, effective development and management, and market trends and consumer behavior by using business intelligence.

Phases of Business Intelligence Life Cycle



Business Intelligence Life Cycle

Phase 1: Analyze Business Requirements

The first step in the Business Intelligence life cycle is to analyze the business requirements. The user identifies the business requirements in order to determine the type of analysis that the user then needs to perform. Identifying the requirements, let the user decides the further action to be performed.

For example, any retail company can analyze the sales data to figure out the products that are top-selling and the products that least sell.

Phase 2: Design Data Model

Once the requirements are identified the user needs to design the logical model according to the requirements. This logical model helps the user to analyze the relationships that exist within the data entities.

For example, For any retail company, the data model consists of products, their customers, and the sales data

Phase 3: Design the Physical Schema

Once the logical model is prepared the next step is to design the physical schema using the data model. The physical schema describes the structure and the content of the data warehouse.

For example, in any retail company, physical schema consists of sales-related facts, product-customer relationships, and the sales transactions

Phase 4: Build the Data Warehouse

Once the logical and physical schema is designed, the next step is to build the data warehouse. The design of a data warehouse depends on the physical and logical schema. After the design of the data warehouse, the data and the content from the source system are loaded into the data warehouse for further steps.

For example, for the retail system, designing the data warehouse consists of developing a database that would store the details of customers, products, and other requirements for the business.

Phase 5: Create the Project Structure (Metadata)

The next step after designing the data warehouse is to create a project structure also known as metadata. With the help of this created project structure, the mapping of the tables and data in the data warehouse is easier. Creating the project structure describes the further steps and types that need to be implemented.

For example, The project structure of the retail company consists of the attributes of the data, the design, and the working flow of the system. This project structure or metadata gives a brief idea about the working of the system.

Phase 6: Develop The BI Objects

The next step is to develop the BI objects such as metrics, attributes, dashboards, reports, and facts. This step consists of developing the reports and dashboards that can be used to analyze the data in the data warehouse.

For example, the retail company can develop reports and statistics charts that can describe the profit and loss margins.

Phase 7: Administer and Maintain the Project

The last step is to administer and maintain the project continuously as it undergoes changes. The project needs to be monitored to maintain the changes, security, and performance of the system.

For example, the retail company needs to monitor the reports and statistics accordingly to increase the profit of the sales.

Conclusion

Business Intelligence provides the proper approach for the data analysis and decision-making process. It helps the business to achieve goals and gain profits. The overall phases of business intelligence make the further development and maintenance process easy.

What Is the Value of Business Intelligence?

Business intelligence's highest value is its ability to support data-driven decisions. BI transforms pools of raw data into useful information that informs decisions and leads to actions that yield positive bottom-line impact.

BI systems drive decisions based on historical, current and potential future data.

- **Descriptive analytics:**

These analytics reveal what has happened or is happening and are part of dashboards, business reporting, data warehousing and scorecards. When managed well, you'll have a better understanding of problem areas in your business and can find opportunities to improve.

- **Predictive analytics:**

These advanced analytics use data mining, predictive modeling, and machine learning to help make projections of future events and assess the likelihood that something will happen.

- **Prescriptive analytics:**

These analytics reveal why you should take a particular action. Prescriptive analytics enable optimization, simulation, decision modeling and provide the best possible analysis for business decisions and actions.

BI software gathers sales, production, financial and many other business data sources. Many companies use industry data to benchmark performance against competitors.

The Benefits of Business Intelligence

Benefit	Description
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Benefit	Description
Visualization	Advanced interactive dashboard representations of data using simple user interfaces offer the ability to visualize information in a graphical format to understand data more insightfully.
Connection	The ability to manage and meld access to various data sources provides a 360-degree view of your business and your company that is not possible in a siloed data environment.
Collaboration	Tools enable data-informed improvements in various business functions like marketing, finance, sales, operations, finance, support, HR and customer care individually and together.
Multi-Platform, Multi-User	BI applications work online and in mobile environments. Tools improve system performance so enterprises can distribute more information to targeted users faster. In multi-terabyte data warehouses, these tools provide excellent query performance.
Scalability	Many systems offer user scalability to support advanced reporting and analysis. Dashboards and reports are available to many users, not just restricted to the organization's data analysts or executives.
Speed and Competitive Edge	BI can perform faster reporting, analysis and planning because of access to global data. The system's analysis capabilities make it possible to react to market or other conditions quickly.
Trusted Data and Accuracy	Reports can be highly customized, and KPIs monitored using more than one data source. Real-time generated reports offer relevant data, which helps organizations, and their employees make better decisions. These reports provide insights, access, accuracy, and relevance.

Benefit	Description
Analysis and Insights	BI processes vast amounts of data to forecast, budget, plan, and stay current. Competitive analysis helps companies understand the competition and benchmark competitor performance. This business intelligence enables product and service differentiation.
Decision-Making Support	Companies gain a competitive edge when they can leverage the existing data at the right time to make accurate decisions faster.
Efficiency and Productivity	A 360-degree view of all activities helps companies identify issues, improve operations, increase sales, and in turn, increase revenue.
Customer Satisfaction	BI can help you identify what services or products you're lacking and improve customer satisfaction by making necessary changes. Reports help you understand customer behavior, develop user personas, and use real-time data on the customer's feedback to make corrective changes and improve customer service and, therefore, satisfaction.
Employee Satisfaction	Using BI data, you can assess team members' strengths and weaknesses and assign relevant training modules to support success. BI tools can automatically recognize positive behavior while regularly tracking worker contributions and improvement.
Savings	BI insight into the corporation's raw data will help decision-makers analyze cost-saving opportunities like excess inventory, human resource redundancies, marketing overages, too many vendors or waste in facilities management.
Savings and Profitability	BI tools can analyze any discrepancies, inefficiencies, or errors. BI helps to increase profit margins by providing insights that lead to future sales and guide where to spend future budgets.

Benefit	Description
Strategic and KPI Targeting	BI assists companies in gaining a competitive edge by helping them find new opportunities and build smarter strategies. Use the data to identify market trends and help improve profit margins for the company. Reports based on tracking established KPIs ensure the enterprise stays on course to match or exceed goals.

Case Studies: Real-World Examples of Business Intelligence at Work

1. Lotte.com: BI Increases Company Revenue

Lotte.com is the leading internet shopping mall in Korea with 13 million customers.

- **Challenge:** With more than 1 million site visitors daily, company executives wanted to understand why customers abandon shopping carts.
- **Solution:** The assistant general manager of the marketing planning team implemented customer experience analytics, the first online behavioral analysis system applied in Korea. The manager used the information to understand customer behavior and implement targeted marketing and transform the website.
- **Results:** With the insights from the new BI analytics program, there was an increase in customer loyalty after one year and an increase of \$10 million in sales. The changes came from identifying the causes of shopping cart abandonment, such as a long checkout process and unexpected delivery times and remedying the situation.

2. Cementos Argos: BI Improves Financial Efficiency

Cementos Argos is a cement company with operations in the U.S., Central and South America and the Caribbean.

- **Challenge:** The company looked for an overall competitive advantage and a way to support better decision-making.
- **Solution:** Cementos Argos created a dedicated business analytics center. The company invested in experienced business analysts and data science teams and used BI to leverage data.
- **Results:** The company standardized the finance process and applied big data to gain more in-depth insight into customer behavior which yielded a higher profitability level.

3. Baylis & Harding: BI Provides Decision Making Process Support

Baylis & Harding is a wholesale distributor specializing in world-class toiletries and gift sets found in major and independent resellers.

- **Challenge:** The company needed to give managers and executives greater visibility into financial, customer and sales data to make better decisions and expand the business.
- **Solution:** Managers and executives used business intelligence tools to create standard and ad hoc reports.
- **Results:** Company executives and managers now have instant access to the business data they need to act proactively. They can create custom dashboards with KPIs relevant to their areas of focus and share the goals and performance details with their teams without having to request a custom report from IT.

Popular Business Intelligence Tools

BI Tool	Description	Key Features	Platform	Strengths
Tableau	A data visualization and business intelligence tool that allows users to connect to various data sources, create interactive dashboards, and share insights.	Data visualization, dashboard creation, data connection, sharing insights.	Desktop, Cloud	Excellent visualizations, user-friendly, strong community support.
Microsoft Power BI	A cloud-based BI tool that allows users to connect to a wide range of data sources, create visualizations, and communicate findings.	Data connection, visualization, cloud service integration, sharing reports.	Cloud, Desktop	Integrates well with Microsoft products, affordable, large user base.

BI Tool	Description	Key Features	Platform	Strengths
QlikView	A platform for data analysis and visualization that helps users create interactive dashboards and explore data in different ways.	Data exploration, interactive dashboards, associative data engine.	Desktop, Cloud	Powerful associative model, strong for self-service BI, great data exploration features.
SAP BusinessObjects	A full BI suite that includes tools for data visualization, reporting, and analytics.	Reporting, analytics, data visualization, integration with SAP data sources.	On-premises, Cloud	Comprehensive suite, strong enterprise support, works well with SAP systems.
IBM Cognos	A BI tool for performance management and corporate intelligence that helps build reports, dashboards, and scorecards.	Reporting, dashboards, performance management, analytics, integration with enterprise systems.	Cloud, On-premises	Excellent for large organizations, strong governance features, customizable.
Oracle Business Intelligence	A comprehensive BI suite with tools for reporting, analytics, and data visualization.	Data visualization, reporting, analytics, integration with Oracle databases.	On-premises, Cloud	Powerful reporting and analytics, integrates well with Oracle databases and enterprise systems.
Looker	A BI and data visualization	Data exploration,	Cloud	Good for modern data exploration,

BI Tool	Description	Key Features	Platform	Strengths
	tool designed to create interactive dashboards and explore data in innovative ways.	custom dashboards, integration with multiple data sources, visualization.		excellent for collaborative analysis, integrates with SQL.

✓ What is a Data Warehouse?

A **Data Warehouse (DW)** is a central repository that stores **large volumes of data** collected from **different sources** for **analysis and reporting** purposes.

- It is optimized for **querying and analysis**, not transaction processing.
- It helps in making **strategic decisions** based on historical data.

🏠 Real-life Example (Indian Context):

Imagine a large retail chain like **Reliance Retail** or **DMart**:

- Daily sales from all stores (Ahmedabad, Surat, Mumbai...) are collected.
- This data is stored in the **Data Warehouse**.
- Managers use this to analyze trends like "**Which products sell best during Diwali?**"

🔑 Key Concepts of Data Warehousing

1. ETL (Extract, Transform, Load)

This is the process of moving data into a data warehouse.

- **Extract:** Get data from sources (e.g., MySQL, Excel, APIs).
- **Transform:** Clean and format the data (remove duplicates, convert ₹ to \$).
- **Load:** Store into the warehouse for analysis.

➡ Think of ETL as a **filtering and packing** process for storage.

2. Data Mart

- A smaller version of a data warehouse.
- Focused on a single subject area like **sales**, **marketing**, or **finance**.
- Example: A **Sales Data Mart** for just analyzing customer purchases.

3. OLAP (Online Analytical Processing)

- Used for complex queries, reporting, and analysis.
- Allows slicing, dicing, drilling down, and pivoting data.

 Example: In an education system, OLAP can help analyze:

- "Student performance by subject by semester by gender."

4. Star Schema vs. Snowflake Schema

☆ *Star Schema*

- Central **fact table** (e.g., sales) linked to **dimension tables** (e.g., product, time, store).
- Simple and faster.

❁ *Snowflake Schema*

- Similar, but dimension tables are **normalized** (broken into multiple related tables).
- More complex but saves space.

5. Fact and Dimension Tables

- **Fact Table:** Quantitative data (sales amount, number of items).
- **Dimension Table:** Descriptive data (product name, customer name, location).

 Example:

Fact_Sales Product_ID Store_ID Date_ID Sales_Amount

Dim_Product Product_ID Name Category

6. Data Warehouse Characteristics (Inmon's Definition)

- **Subject-Oriented:** Focuses on a specific subject (sales, finance).
- **Integrated:** Combines data from different sources.
- **Non-Volatile:** Once entered, data doesn't change.
- **Time-Variant:** Data is kept over time (historical data).

7. Tools Used

- **ETL Tools:** Talend, Informatica, Apache Nifi
- **Data Warehouse Tools:** Amazon Redshift, Google BigQuery, Snowflake, Oracle DW
- **BI Tools for Analysis:** Tableau, Power BI, QlikView



Benefits of Data Warehousing

- Better **decision-making**.
- **Historical insights** into data.
- Improved **data quality and consistency**.
- Faster **reporting and analytics**.